

Power BI Assignment -1

Data Import & Transformation

Objective

The objective of this assignment is to demonstrate the process of importing, cleaning, transforming, and modeling data using Power BI. The project also involves creating calculated columns, handling missing values, merging datasets, grouping data, and establishing relationships between tables for effective data analysis.

1. Data Import

Objective

Import multiple datasets into Power BI for analysis.

Steps Performed

- Opened **Power BI Desktop**.
- Selected **Get Data**.
- Chose **Excel Workbook**.
- Imported the following tables:
 - List of Orders
 - Order Details
 - Sales Target
- Clicked **Transform Data** to open Power Query Editor.

Result

All datasets were successfully imported into Power BI and loaded into Power Query Editor for transformation.

2. Row Limiting & Data Types

Objective

Reduce unnecessary records and assign appropriate data types.

Steps Performed

Row Limiting

- Selected the **List of Orders** table.
- Navigated to:

Home → Keep Rows → Keep Top Rows

- Entered **500**.

Only the first **500 rows** were retained.

Data Types

Verified and changed data types as required.

Result

The dataset now contains only the required records with correct data types.

3. Text Formatting

Objective

Standardize text values for consistency.

Steps Performed

Applied text transformations using:

Transform → Format

Operations performed:

Result

Removed extra spaces, hidden characters, and inconsistent capitalization.

4. Column Creation

Objective

Create additional columns required for analysis.

Steps Performed

Created custom columns using **Add Column** → **Custom Column**.

Examples include:

Profit Percentage

```
([Profit] / [Sales]) * 100
```

Sales Category

```
If Sales >=1000 then "High"  
Else if Sales >=500 then "Medium"  
Else "Low"
```

Result

New calculated columns were successfully added.

5. Conditional Column

Objective

Categorize records based on Profit.

Column Name

Profit Status

Steps Performed

Selected

Add Column → Conditional Column

Created the following conditions:

Condition	Output
Profit < 0	Loss
Profit = 0	Break-Even
Profit > 0	Profit

Result

Each order was categorized according to its profit.

6. Merging Data (Joins)

Objective

Combine related datasets.

Merge 1

Merged:

List of Orders

with

Order Details

Using

Order ID

Join Type:

Left Outer Join

Expanded columns:

- Category
 - Sub Category
 - Sales
 - Profit
 - Quantity
-

Merge 2

Merged:

Order Details

with

Sales Target

Using

Category

Join Type:

Left Outer Join

Expanded:

- Target

Result

Successfully combined related information into a unified dataset.

7. Handling Missing Data & Duplicate Data

Missing Data

Objective

Identify and handle null values.

Steps Performed

Used:

Home → Remove Rows

or

Transform → Replace Values

Actions:

- Replaced missing values where appropriate.
 - Removed rows containing critical null values.
-

Duplicate Data

Objective

Remove duplicate records.

Steps Performed

Selected required columns.

Used:

Home → Remove Rows → Remove Duplicates

Result

Dataset became cleaner and free from duplicate records.

8. Sorting and Filtering Data

Objective

Organize and filter records.

Sorting

Applied sorting to:

- Order Date
- Sales
- Profit

Both Ascending and Descending.

Filtering

Applied filters for:

- Category
- Region
- Profit
- State

Used Power Query filter options to display only required records.

Result

The dataset became easier to analyze.

9. Grouping and Aggregating Data

Objective

Summarize data using Group By.

Steps Performed

Selected:

Category

Then:

Transform → Group By

Created the following aggregations:

Result

Data was summarized for category-wise analysis.

10. Data Modeling

Objective

Establish relationships between tables for accurate reporting.

Relationships Created

Table 1	Column	Table 2	Column
List of Orders	Order ID	Order Details	Order ID
Order Details	Category	Sales Target	Category

Relationship Settings

- Cardinality:
 - One-to-Many (where applicable)
- Cross Filter Direction:
 - Single
- Active Relationship:
 - Yes

Steps Performed

- Opened **Model View**.
- Selected **Manage Relationships**.
- Created relationships using the specified columns.
- Verified that relationships were active and correctly defined.

Result

The data model was successfully established, enabling accurate filtering, aggregation, and visualization across related tables.

Conclusion

This Power BI assignment demonstrated the complete **Extract, Transform, Load (ETL)** workflow. Data was imported, cleaned, transformed, enriched with custom and conditional columns, merged across multiple tables, and modeled using relationships. These transformations ensured a high-quality dataset suitable for creating interactive reports and dashboards, supporting accurate business insights and decision-making.